

Abelian Extensions Problem Set 3

1. Show that if p is a prime and G is an abelian group with a unique subgroup of index p then G is cyclic.
2. Let $L = \mathbb{Q}(\sqrt{d})$ where $d \in \mathbb{Z}$ is square free. Using the fact that $\delta_L = d$ if d is 1 (mod 4) and $\delta_L = 4d$ if d is 2 or 3 (mod 4), determine all of the quadratic extensions L for which δ_L is a power of 2.
3. Suppose L/K is an extension of number fields. Show that for any $\sigma \in \text{Aut}(L/K)$ and for any $a \in K$, the ideal of $\mathcal{O}_L$ given by

$$a\mathcal{O}_L = \{a\beta : \beta \in \mathcal{O}_L\}$$

satisfies $\sigma(a\mathcal{O}_L) = a\mathcal{O}_L$.

4. Using the exercise above, show that if $p \in \mathbb{Z}$ factors in $\mathcal{O}_L$ as

$$p\mathcal{O}_L = \mathcal{P}_1^{e_1} \mathcal{P}_2^{e_2} \cdots \mathcal{P}_g^{e_g},$$

then $\sigma(\mathcal{P}_i) = \mathcal{P}_j$ for some $1 \leq j \leq g$.

5. Show that if $\mathcal{P}$ is a prime ideal of $\mathcal{O}_L$ and $p = \mathcal{P} \cap \mathbb{Z}$, then the decomposition group $D(\mathcal{P}|p) = \{\sigma \in \text{Gal}(L/\mathbb{Q}) \mid \sigma(\mathcal{P}) = \mathcal{P}\}$ is a subgroup of G . Show also that the inertia group $I(\mathcal{P}|p) = \{\sigma \in \text{Gal}(L/\mathbb{Q}) \mid \sigma(a) \equiv a \pmod{\mathcal{P}} \text{ for all } a \in \mathcal{O}_L\}$ is a subgroup of $D(\mathcal{P}|p)$.
6. For $\sigma \in D(\mathcal{P}|p)$ define the function $\bar{\sigma} : \mathcal{O}_L/\mathcal{P} \rightarrow \mathcal{O}_L/\mathcal{P}$ by $\bar{\sigma}(a) \equiv \sigma(a) \pmod{\mathcal{P}}$. Show that the map $D(\mathcal{P}|p) \rightarrow \text{Gal}((\mathcal{O}_L/\mathcal{P})/\mathbb{F}_p)$ given by $\sigma \mapsto \bar{\sigma}$ is a group homomorphism with kernel $I(\mathcal{P}|p)$. Show also that if p is unramified in L , then $D(\mathcal{P}|p)$ is cyclic.
7. Recall that for $m \geq 0$, the m th ramification group of $\mathcal{P}$ is

$$V_m = \{\sigma \in G : \sigma(a) \equiv a \pmod{\mathcal{P}^{m+1}} \text{ for all } a \in \mathcal{O}_L\}.$$

Show that $I(\mathcal{P}|p) = V_0 \geq V_1 \geq V_2 \geq \dots$. Show also that for some $m \in \mathbb{Z}$, V_m is the trivial group. Here's a hint for the second part:

Suppose $p \in \mathbb{Z}$ is prime. If you have some integer $n \in \bigcap_{k=1}^{\infty} \mathbb{Z}/p^k\mathbb{Z}$, then what must that integer be? (think about what $n \in p^k\mathbb{Z}$ means). Generalize that idea to $a \in \bigcap_{m=0}^{\infty} \mathcal{O}_L/\mathcal{P}^{m+1}$, and consider what has to be true of $\sigma \in \bigcap_{m=1}^{\infty} V_m$.